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Answer Explanation:

Apply

Let x be the number of robots that can be installed in Year 1. Then

$$\begin{aligned} 600{,}000 + 150{,}000x &\geq 400{,}000x \\ 600{,}000 &\geq 250{,}000x \\ \frac{600{,}000}{250{,}000} &\geq x \\ 2.4 &\geq x \end{aligned}$$

The maximum number of robots that can be installed in Year 1 is 2. The budget remaining to be carried over to Year 2 is 600,000 + 2(150,000) − 2(400,000) = 100,000.

Assuming that at the beginning of Year 2 there are 2 robots working, let t be the number of robots that can be installed in Year 2. Then

$$\begin{aligned} 100{,}000 + 2(150{,}000) + 150{,}000t &\geq 400{,}000t \\ 400{,}000 &\geq 250{,}000t \\ \frac{400{,}000}{250{,}000} &\geq t \\ 1.6 &\geq t \end{aligned}$$

The maximum number of robots that can be installed in Year 2 is 1. The budget remaining to be carried over to Year 3 is 100,000 + 3(150,000) − 400,000 = 150,000.

Assuming that at the beginning of Year 3 there are 2 + 1 = 3 robots working, let y be the number of robots that can be installed in Year 3. Then

$$\begin{aligned} 150{,}000 + 3(150{,}000) + 150{,}000y &\geq 400{,}00y \\ 600{,}000 &\geq 250{,}000y \\ \frac{600{,}000}{250{,}000} &\geq y \\ 2.4 &\geq y \end{aligned}$$

The maximum number of robots that can be installed in Year 3 is 2.

***X*:**

From the explanation above, the maximum number of robots that can be installed in Year 1 is 2.

The correct answer is 2.

***Y*:**

From the explanation above, the maximum numbers of robots that can be installed in Years 1, 2, and 3, respectively, are 2, 1, and 2. The maximum number of robots that can be in service in Year 3 is 2 + 1 + 2 = 5.

The correct answer is 5.

•

While designing a game involving chance, Desmond noticed that the probability a fair coin lands face up exactly 2 times when the coin is tossed 3 times is equal to the probability that a fair coin lands face up exactly m times when the coin is tossed n times, where $n > 3$.

Select for m and for n values consistent with the given information. Make only two selections, one in each column.

m	n	

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		1
		2
		3
		4
		5
		6

Answer Explanation:

Infer

m**:**

If a coin is tossed 3 times and each toss can land only one of face up (U) or face down (D), then there are 3 possible outcomes where the coin lands face up exactly 2 times: UUD, UDU, and DUU. Here the notation is such that UDU, for example, means the coin landed face up, face down, and face up for the first, second, and third tosses, respectively. Since the coin is fair, for each toss, the probability of landing face up is $\frac{1}{2}$ and the probability of landing face down is $\frac{1}{2}$. Therefore, the probability of each of the possibilities UUD, UDU, and DUU is $\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^3$, and the probability that one of these three possibilities occurs is $3 \cdot \left(\frac{1}{2}\right)^3 = 38$.

If the coin is tossed 4 times (i.e., $n = 4$), then the probability of each of the possibilities UUUU, UUUD, UUDU,…, UDDU, and so on is $\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^4$, and the probability that the coin lands face up exactly m times for $m = 0, 1, 2, 3$, and 4 is given in the following table.

m = number of times face up	possible outcomes	number of possible outcomes	probability
0	DDDD	1	$1 \cdot \left(\frac{1}{2}\right)^4$
1	DDDU, DDUD, DUDD, UDDD	4	$4 \cdot \left(\frac{1}{2}\right)^4$
2	DDUU, DUDU, DUUD, UUDD, UDUD, UDDU	6	$6 \cdot \left(\frac{1}{2}\right)^4$
3	UUUD, UUDU, UDUU,	4	$4 \cdot \left(\frac{1}{2}\right)^4$

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	DUUU		
4	UUUU	1	1·(12)4

Since $6 \cdot \left(\frac{1}{2}\right)^4 = \frac{3}{8}$, the correct selection of values is $m = 2$ and $n = 4$.

The correct answer is *2*.

***n*:**

The explanation above shows that $n = 4$ for the correct selection of values of m and n.

The correct answer is *4*.

•

Linguist: Plosives and fricatives are two classes of consonants. A "voicing contrast" is a distinction between two consonants that are identical except that one is voiced and the other is unvoiced. In language family X, languages with voicing contrasts in their fricatives always have voicing contrasts in their plosives. This means that in that family, any given language has a voicing contrast in its fricatives _____ it has a voicing contrast in its plosives. In other words, a given language in that family lacks any voicing contrasts in its plosives _____ it lacks any such contrasts in its fricatives.

Select for *First blank* the word or phrase that most logically completes the statement with the first blank. And select for *Second blank* the word or phrase that most logically completes the statement with the second blank. Make only two selections, one in each column.

First blank	Second blank	
		and
		if
		only if
		or
		unless

Answer Explanation:

Recognize

***First blank*:**

Letting P denote "has a voicing contrast in its fricatives" and Q denote "has a voicing contrast in its plosives," then the first identified sentence states that given P, we have Q, or "if P, then Q," which is logically equivalent to "P only if Q."

The correct answer is *only if*.

***Second blank*:**

Letting P and Q be as above, we are to put the second identified sentence into the form "not-Q _____ not-P." Since the contrapositive of the first identified sentence is "if not-Q, then not-P," which is logically equivalent to the first identified sentence, it follows that a logically equivalent restatement of the first identified sentence is "not-Q only if not-P."

The correct answer is *only if*.

•

At a certain university, there is a strong positive correlation between the time of day at which university classes are

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At a certain university, there is a strong positive correlation between the time of day at which university classes are offered and the classes' average (arithmetic mean) grades, with earlier times associated with lower grades. However, for the classes offered at the earliest time but not for any other classes, there was a strong negative correlation between the grades of students in those classes and the number of scheduled classes they missed. In fact, when the grades of students who missed at least 5 scheduled classes were excluded, classes offered at the earliest time more often than not had significantly higher average grades than classes offered at any other time.

Consider the following statement:

At the university in question, classes offered at earlier times ____1____ lower average grades than classes offered later in the day, but when the grades of students who missed at least 5 scheduled classes were excluded, classes offered at the earliest time____2____ higher average grades than classes offered later in the day.

Select for *1* and for *2* the options that complete the sentence so that it most accurately summarizes the information provided. Make only two selections, one in each column.

1	2	
		tended to have
		tended not to have
		almost always had
		seldom, if ever, had
		possibly had

Answer Explanation:

Recognize

***1*:**

Because there is a strong positive correlation between the time of day and class average, earlier times tend to correspond to lower class averages. Thus, it is clear that none of the second, fourth, or fifth answer choices correctly completes the sentence. Also, the following argument shows that the use of the third answer choice less accurately summarizes the information provided than the use of the first answer choice. To see that the third answer choice may not accurately convey that there is a strong positive correlation between time of day and class average, note that a strong positive correlation can exist between two equal-sized collections of data values without almost all of the data values in one collection being in the same numerical order as the corresponding data values in the other collection. This can occur, for example, when relatively few of the data values in each collection are extremes and have the same corresponding order and relatively many of the data values in each collection are nearly equal and have the opposite corresponding order.

The correct answer is *tended to have*.

***2*:**

According to the information provided, when grades of students who missed at least 5 scheduled classes are excluded, there is a strong negative correlation between the grades for classes offered at the earliest time and grades for classes offered at later times. Therefore, with this exclusion, grades for the earliest time tend to be higher than grades for later times. Thus, it is clear that none of the second, fourth, or fifth answer choices correctly completes the sentence. Also, an argument similar to that given above shows that the use of the third answer choice less accurately summarizes the information provided than the use of the first answer choice.

The correct answer is *tended to have*.

•

Each Monday through Friday (a *workweek*), Avinash will bring either exactly one apple or exactly one banana with him to his workplace for an afternoon snack. To avoid having to decide which to bring, each morning Avinash will toss a coin with a face on exactly one side that is equally likely to land face up or face down. If the coin lands face up, then he will bring an apple, and if the coin lands face down, then he will bring a banana. Avinash correctly

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then he will bring an apple; and if the coin lands face down, then he will bring a banana. Avinash correctly determined the probability that, for a given workweek, either he would bring an apple on at least 4 consecutive days or he would bring a banana on at least 4 consecutive days. This probability was m divided by n.

Select for m and for n values jointly consistent with the given information. Make only two selections, one in each column.

m	n	
		4
		6
		8
		12
		20
		32

Answer Explanation:

Infer

m:

In a given workweek, Avinash will toss the coin 5 times. There are 6 possible outcomes from tossing the coin 5 times that result in Avinash bringing an apple on at least 4 consecutive days or bringing a banana on at least 4 consecutive days: UUUUD, DUUUU, UUUUU, DDDDU, UDDDD, and DDDDD. Here the notation is such that DDDDU, for example, means the coin landed face down from Monday through Thursday and face up on Friday. Since the coin is equally likely to land face up or face down, for each of the 5 days in the workweek, the probability is $\frac{1}{2}$ that Avinash will bring an apple, and the probability is $\frac{1}{2}$ that Avinash will bring a banana. Therefore, the probability of each of these possibilities is $\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^5$, and the probability that one of these 6 possibilities occurs is $6 \cdot \left(\frac{1}{2}\right)^5 = \frac{6}{32}$. The correct selection of values is $m = 6$ and $n = 32$.

The correct answer is *6*.

n:

The explanation above shows that $n = 32$ for the correct selection of values of m and n.

The correct answer is *32*.

•

Philosophy student: Some objects that are considered beautiful by everyone who has observed them may not be, in fact, truly beautiful. To see that this is so, consider this: No one doubts that some objects that are appreciated by many people have aesthetic flaws that are discernible only to sophisticated observers. But even these sophisticated observers are limited by their finite intellects and experiences. Thus, an object that appears beautiful to the most sophisticated actual observers may nonetheless have subtle but severe aesthetic shortcomings that would make it appear hideous to hypothetical observers of even greater sophistication. Such an object would be ugly, regardless of any actual person's opinion.

In general, if an object ____1____, then that object ____2____.

Select for *1* and for *2* the two different options that complete the sentence in such a way that it expresses a principle on which the philosophy student's argument relies. Make only two selections, one in each column.

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on which the philosophy student's argument relies. Make only two selections, one in each column.

1	2	
		is considered beautiful by everyone
		is thought by most observers to have some aesthetic flaws
		would appear hideous to hypothetical observers of even greater sophistication than the most sophisticated actual observers
		is not truly beautiful
		is not widely appreciated by unsophisticated observers

Answer Explanation:

Evaluate

The philosophy student's argument is based on extrapolating from a claim that the student asserts everyone agrees with, namely, that there are art objects that many people appreciate but that have aesthetic flaws that only more sophisticated observers can discern. The student notes that even the most sophisticated observers have finite intellects and experiences.

The student then suggests that because of this, there may be art objects that sophisticated viewers appreciate but nonetheless have aesthetic flaws that a hypothetical even-more-sophisticated observer might discern and, as a result, would find the art object ugly. The student draws the conclusion that such an object would in fact be ugly, even if no actual person discerns the flaw.

Finally, the student infers from all this that some objects that everyone considers beautiful may not be truly beautiful. To answer this question correctly, we must construct a sentence that expresses a principle that underlies the student's argument.

Note that the student's argument moves from an assertion about hypothetic observers with greater sophistication than any actual observers who find an art object hideous to a claim about art objects that may not truly be beautiful even though all actual observers think the object is beautiful. The student must be assuming that there is a link between hypothetical sophisticated viewers finding an object hideous and the object not being truly beautiful.

Therefore, the student's argument seems to rely on the principle that, in general, if an object would appear hideous to hypothetical observers of even greater sophistication than the most sophisticated actual observers, then that object is not truly beautiful.

***1*:**

The correct answer is *would appear hideous to hypothetical observers of even greater sophistication than the most sophisticated actual observers.*

***2*:**

The correct answer is *not truly beautiful.*

•

The following statements describe certain characteristics of a certain pool of candidates for a position. Any candidate who did not meet the minimum qualifications for the position was immediately excluded from consideration. The two candidates who met the minimum qualifications for the position and met all of the desired qualifications also had multiple recommendations. All candidates who received a telephone interview also had extensive experience. All candidates who had extensive experience and impressed the hiring committee during the telephone interview were invited to interview on-site. At least one candidate declined an invitation for an on-site interview, and exactly one candidate was interviewed on-site without receiving a telephone interview.

Consider the following incomplete sentence:

If any candidate ____1____, then that candidate ____2____.

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Select for *1* and for *2* two different options that best complete the sentence such that it can be logically inferred from the information provided. Make only two selections, one in each column.

1	2	
		did not meet the minimum qualifications
		had multiple recommendations
		had extensive experience
		impressed the hiring committee during the telephone interview
		interviewed on-site

Answer Explanation:

Infer

If any candidate impressed the hiring committee during the telephone interview, then it follows that the candidate had a telephone interview, and hence (directly by one of the statements) the candidate had extensive experience. Therefore, if any candidate impressed the hiring committee during the telephone interview, then that candidate had extensive experience.

***1*:**

The correct answer is *impressed the hiring committee during the telephone interview*.

***2*:**

The correct answer is *had extensive experience*.

•

On a 12-hour analog clock, the hour hand moves at a constant rate of 1 revolution every 12 hours, and the minute hand moves at a constant rate of 1 revolution every hour. The hands are perpendicular at 3:00 in the morning. To the nearest second, the next time they are superimposed (i.e., both pointing at the same point on the outer rim of the clock face) is M minutes and S seconds after 3:00 in the morning.

Select for M and for S values that are consistent with the information provided. Make only two selections, one in each column.

M	S	
		15
		16
		17
		20
		22
		34

Answer Explanation:

Infer

***M*:**

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***M*:**

For convenience, label the points on the outer rim of the clock by "rim-minutes" (rm), where, for example, the point corresponding to 1:00 is 5 rm, the point corresponding to 2:00 is 10 rm, the point corresponding to 3:00 corresponds to 15 rm, and so on. Let t denote the number of minutes of time after 3:00 when the minute hand and hour hand are next superimposed. In t minutes after 3:00, the minute hand, which was initially pointing at 0 rm and travels at a rate of 1 rm/min, will be pointing at (t min)(1 rm/min) + 0 rm = t rm. Also, in t minutes after 3:00, the hour hand, which was initially pointing at 15 rm and travels at a rate that is $\frac{1}{12}$ the rate of the minute hand (i.e., travels at a rate of 112 rm/min), will be pointing at (t min)$\left(\frac{1}{12}\ \mathrm{rm/min}\right) + 15\ \mathrm{rm} = \left(\frac{1}{12}t + 15\right)$ rm. Since the minute hand and the hour hand will be pointing at the same point on the rim t minutes after 3:00, it follows that $t = \frac{1}{12}t + 15$, or $t = \frac{180}{11} \approx 16.36$ min.

Therefore, the desired time is 16 minutes and a certain number (less than 60) of seconds.

The correct answer is *16*.

***S*:**

Because $\frac{180}{11} = 16 + \frac{4}{11}$, and $\frac{4}{11}$ minutes equals $60\left(\frac{4}{11}\right) \approx 21.818$ seconds, to the nearest second the desired time after 3:00 is 16 minutes and 22 seconds.

The correct answer is *22*.

•

The following argument is logically flawed. The author's goal was to craft the argument so that the conclusion follows logically from Premises 1 and 2 and so that both premises are necessary to draw the conclusion.

Premise 1: Every respondent to our survey who **reported feeling satisfied** also reported being in a good mood.

Premise 2: Every respondent to our survey who **reported having a central goal** also **reported being in a good mood**.

Conclusion: Therefore, assuming all of the reports were accurate and complete, every respondent to our survey who **felt satisfied** also **had a central goal**.

Select for *Boldface A* and for *Boldface B* two of the boldfaced phrases in the argument such that Boldface A occurs earlier in the argument than Boldface B, and exchanging the positions of those two phrases in the argument would make it so the argument fulfills the author's goal. Make only two selections, one in each column.

Boldface A	Boldface B	
		reported feeling satisfied
		reported having a central goal
		reported being in a good mood
		felt satisfied
		had a central goal

Answer Explanation:

Strategize

A simple example involving only two respondents can be used to show the original argument is logically flawed. Assume only Respondent X felt satisfied, only Respondent Y had a central goal, and only Respondents X and Y were

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Assume only Respondent X felt satisfied, only Respondent Y had a central goal, and only Respondents X and Y were in a good mood. In this example, the two original premises are true, but the conclusion is false because Respondent X felt satisfied and did not have a central goal.

If the positions of the second and third boldfaced phrases are exchanged, then the revised argument is the following, in which the conclusion logically follows from both premises and both premises are necessary.

Premise 1: Every respondent **feeling satisfied** was in a **good mood.**

Premise 2: Every respondent in a **good mood** has a **central goal.**

Conclusion: Every respondent **feeling satisfied** has a **central goal.**

***Boldface A*:**

The correct answer is *reported having a central goal.*

***Boldface B*:**

The correct answer is *reported being in a good mood.*

•

In Country C, some but not all eligible voters are required to vote. The particulars of the country's laws governing voting are as follows:

Every citizen who is eligible must vote on election day.

A person is eligible if (and only if) he or she meets the age requirement and either is a citizen or meets the residency requirement for noncitizens.

The age requirement is that every voter must be at least 19 years old on election day.

The residency requirement for noncitizens is that the voter must have been a resident of Country C for at least 5 years on election day.

Consider the following individuals:

Abigail: a citizen who is currently 19 years old

Barbara: a 7-year resident noncitizen who is currently 19 years old

Charles: a 7-year resident noncitizen who is currently 18 years old

For an election held today, select the individual or individuals who must vote, based on the information provided, and select the individual or individuals who must not vote, based on the information provided. Make only two selections, one in each column.

Must vote	Must not vote	
		Abigail only
		Barbara only
		Charles only
		Abigail and Barbara only
		Abigail and Charles only
		Barbara and Charles only

Answer Explanation:

Recognize

***Must vote*:**

The information given states that in Country C, some eligible voters, but not all, are required to vote. Among the particulars of the law governing voting for Country C, there is only one regarding who must vote, namely the first

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particulars of the laws governing voting for Country C, there is only one regarding who *must vote*, namely the first particular listed, which states that every citizen who is eligible must vote on election day. Of the three individuals under consideration—Abigail, Barbara, and Charles—only Abigail is a citizen, and Abigail is currently 19, which is the minimum age for voting. Given that every citizen who meets the age requirement is eligible and—as we've seen—every eligible citizen must vote, then for an election held today in Country C, Abigail must vote, and of the three individuals under consideration, Abigail is the only one of them who *must vote*.

The correct answer is *Abigail only*.

***Must not vote*:**

As discussed above, Abigail *must vote*, so we can immediately rule out each of the responses that include Abigail because if she *must vote*, then it cannot be the case that she *must not vote*. What about Barbara and Charles? Barbara is currently 19 years old and is a 7-year resident noncitizen; Charles is currently 18 years old and is also a 7-year resident noncitizen. The particulars listed indicate that a voter is eligible if, and only if, he or she meets the age requirement, which is 19, and either is a citizen or meets the residency requirement for noncitizens, which is a minimum of 5 years. Both Barbara and Charles meet the residency requirement for eligibility, but only Barbara meets the age requirement. Therefore, although Barbara is eligible, Charles is not eligible, and therefore he, and only he among the three individuals under consideration, *must not vote* in any election held today in Country C.

The correct answer is *Charles only*.

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